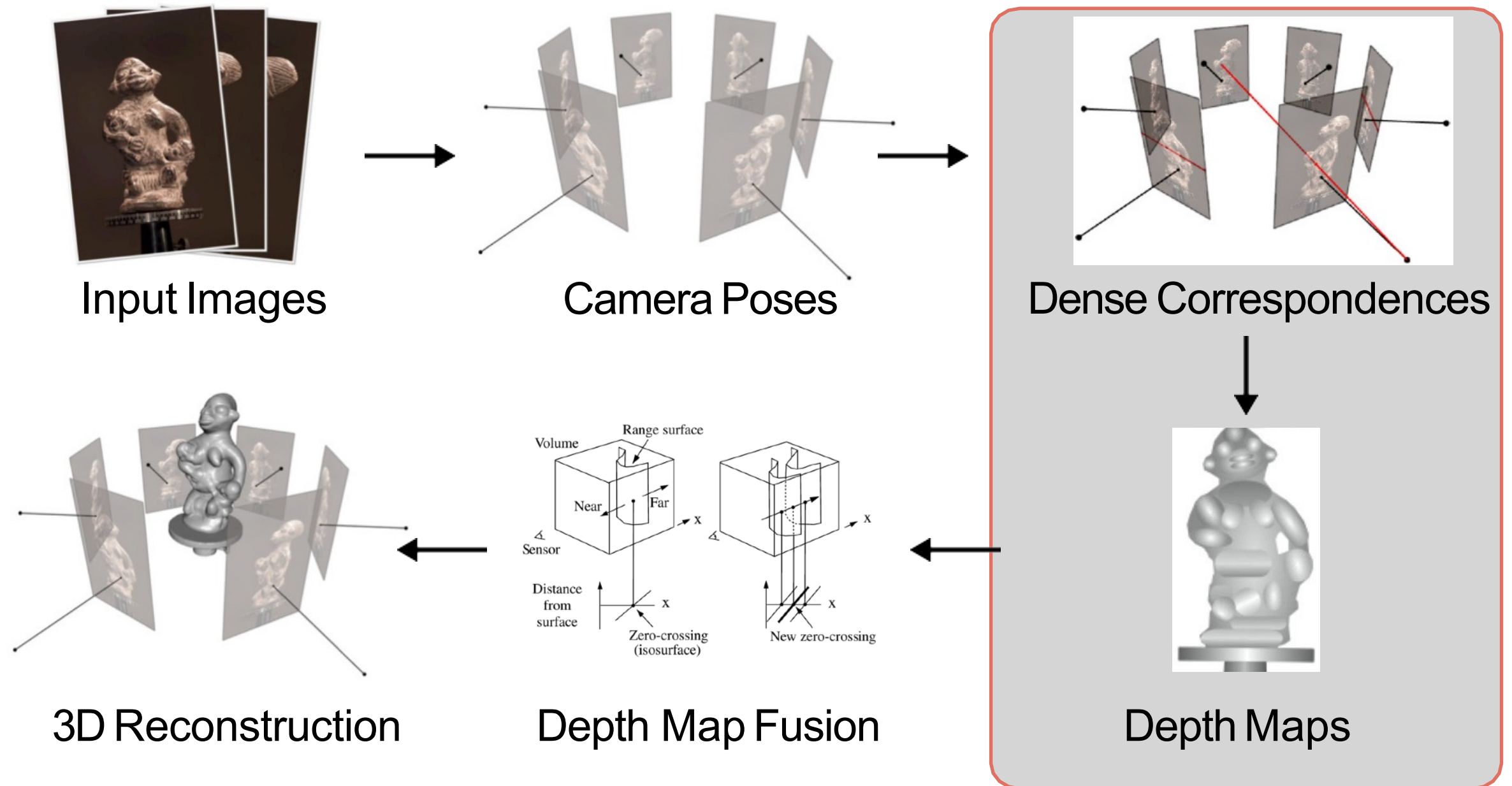


Recap

3D Reconstruction Pipeline



The fundamental matrix $\mathbf{F}$

Consider the essential matrix

$$\hat{\mathbf{x}}'^{\top} \mathbf{E} \hat{\mathbf{x}} = 0$$

Construct the camera
coordinate points from
pixels using the intrinsics

K = intrinsics

$$\hat{\mathbf{x}}' = \mathbf{K}'^{-1} \mathbf{x}' \qquad \hat{\mathbf{x}} = \mathbf{K}^{-1} \mathbf{x}$$

Rewrite

$$\mathbf{K}'^{-\top} \mathbf{E} \mathbf{K}^{-1} \mathbf{x} = 0$$
$$\mathbf{x}'^{\top} (\mathbf{K}'^{-\top} \mathbf{E} \mathbf{K}^{-1}) \mathbf{x} = 0$$

$$\mathbf{x}'^{\top} \mathbf{F} \mathbf{x} = 0$$

The 8 point algorithm

- Normalize the point correspondences
- Construct the $M \times 9$ design matrix $\mathbf{A}$
- Computer the singular value decomposition of $\mathbf{A}$
- Find elements of $\mathbf{F}$ as the coefficients of the column in $\mathbf{V}$ corresponding to the smaller Eigenvalue
- Enforce rank constraint $r = 2$
- Un-normalize $\mathbf{F}$

Rectification: Example



- Correspondences are located on the **same image row** as the querypoint

Disparity estimation



- The relative horizontal displacement is called **disparity** (here: 0-160 px to the left)
- Disparity is anti-proportional to depth (far points move less than closer points)

Disparity to depth

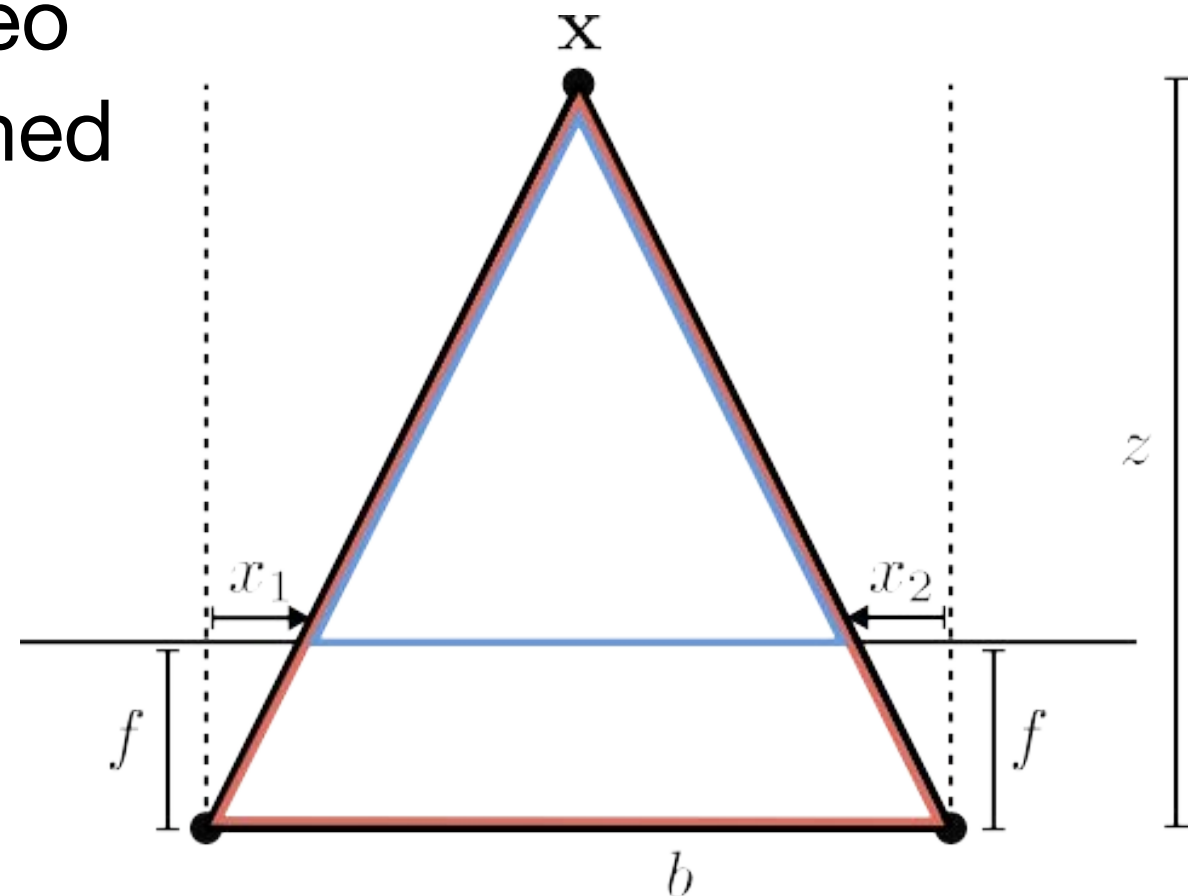
The left and right ray intersect
(shared epipolar plane).

Assume the disparity is $d = x_1 - x_2$.
Given a focal length f and the stereo
baseline b , the depth z is constrained
by

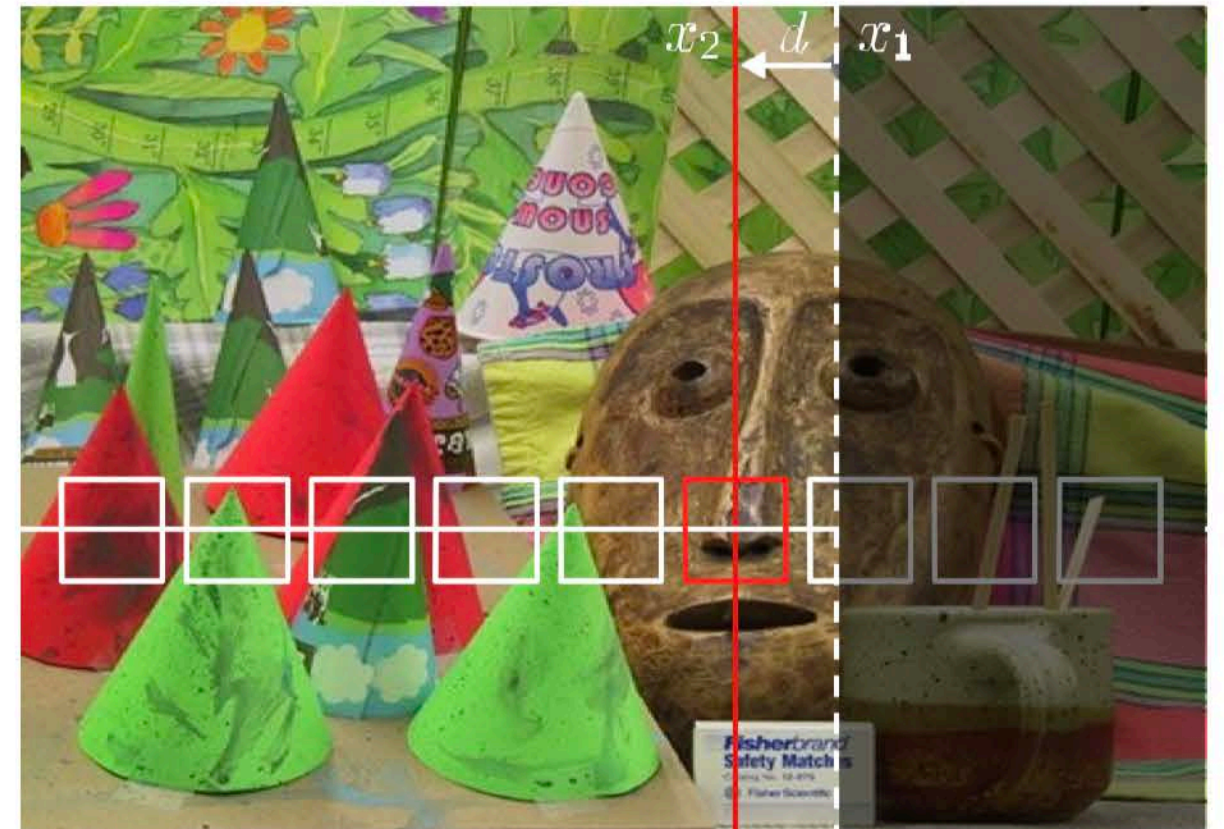
$$\frac{z - f}{b - d} = \frac{z}{b}$$

$$zb - fb = zb - zd$$

$$z = \frac{fb}{d}$$

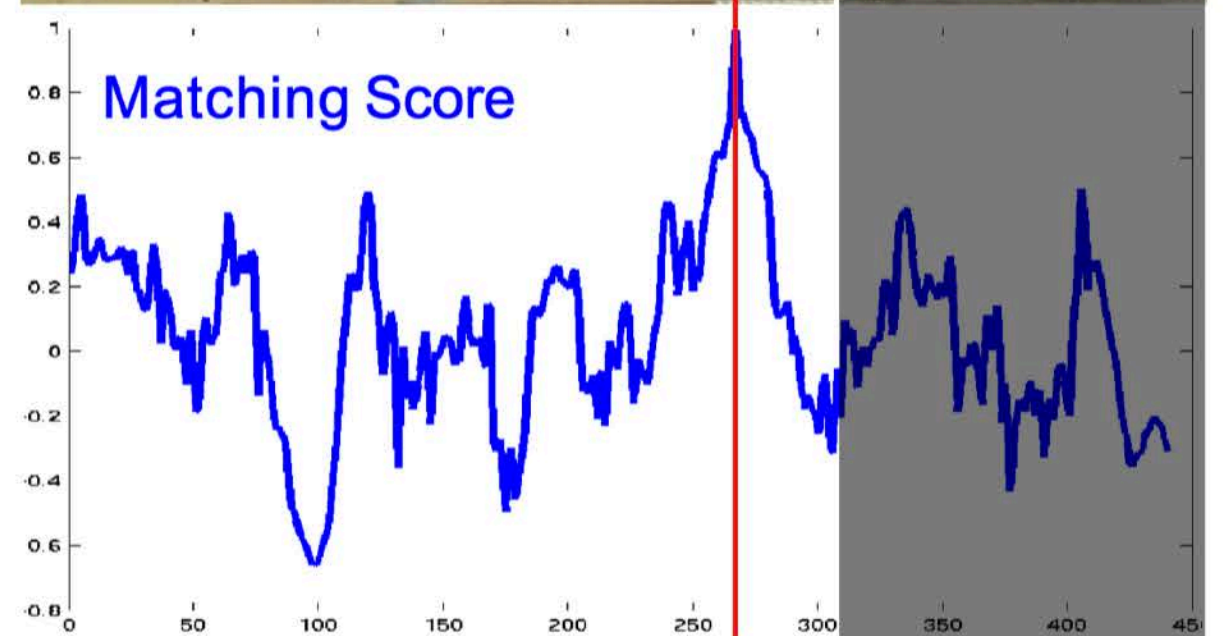


Stereo matching

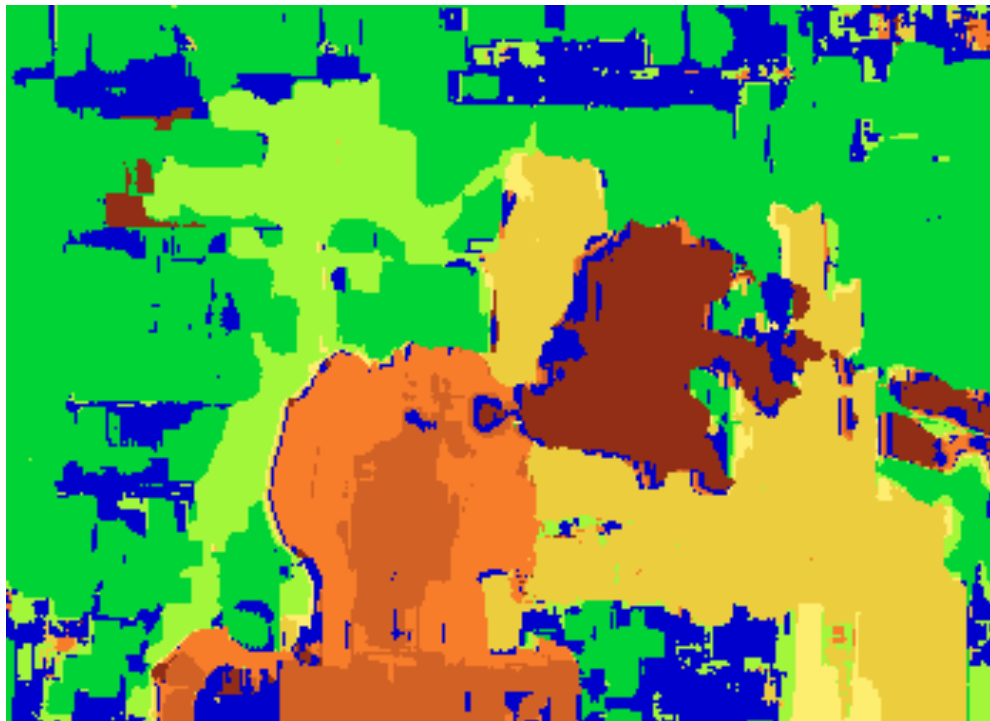


Block matching

- Choose disparity range
- For all pixels, compute best disparity
- Repeat in opposite direction, remove inconsistencies to reduce outliers



Block matching result



Window-based matching
(best window size)



Ground truth

MRF result (via graph cuts)



Graph cuts-based method



Ground truth

Boykov et al., [Fast Approximate Energy Minimization via Graph Cuts](#),

International Conference on Computer Vision 1999.

Summary

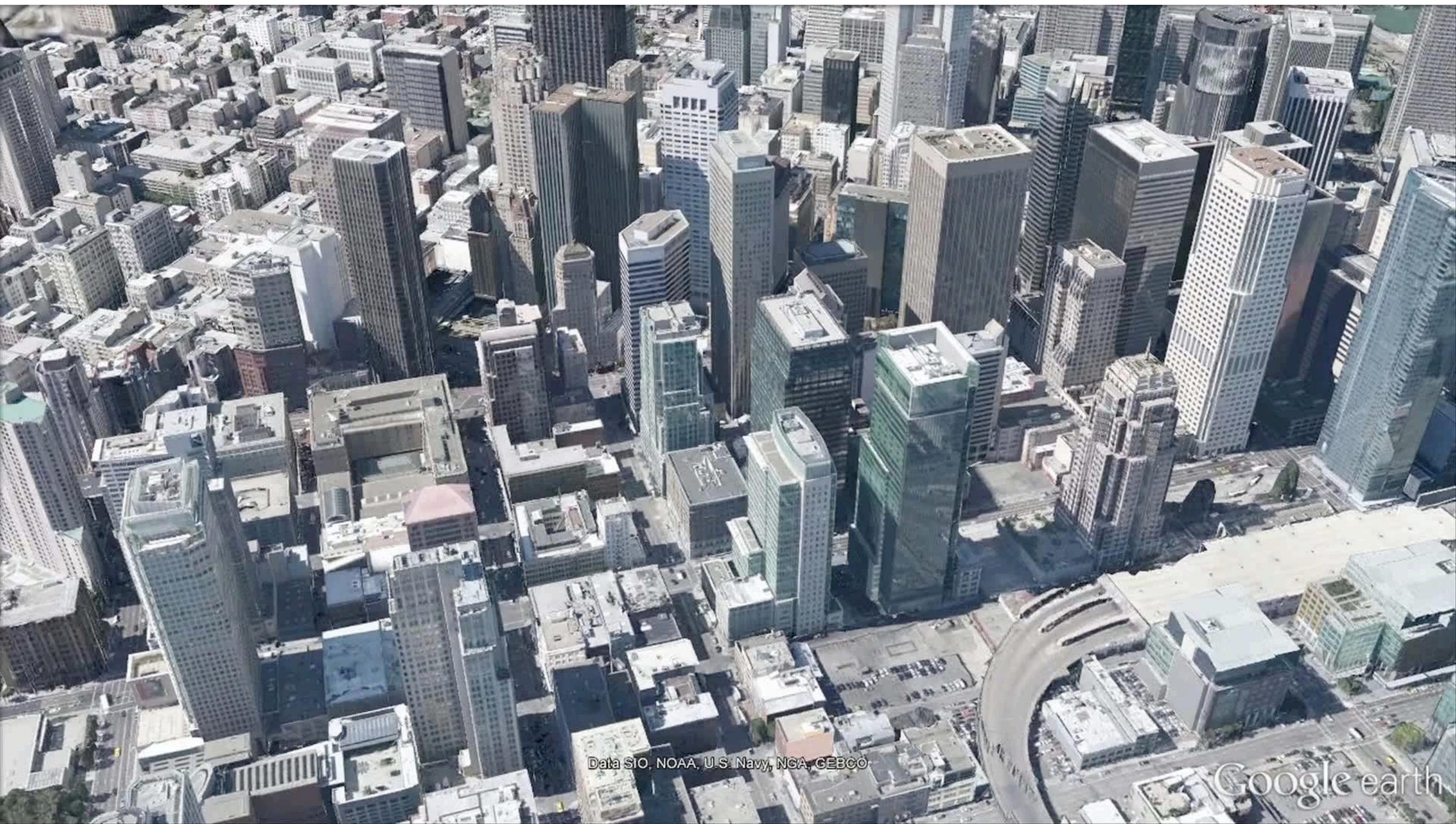
- ▶ Block matching suffers from **ambiguities**
- ▶ Choosing window size is problematic (tradeoff)
- ▶ Incorporating **smoothness constraints** can resolve some of the ambiguities and allows for choosing small windows (no bleeding artifacts)
- ▶ Can be formulated as **MAP inference in a discrete MRF**
- ▶ MAP solution can be obtained using belief propagation, graph cuts, etc.
- ▶ Integrating recognition cues can further regularize the problem



RED Lion movie shorts

https://www.youtube.com/watch?v=FOitGa117_s





Data SIO, NOAA, U.S. Navy, NGA, GEBCO

Google earth

Whistle in the Form of Female Figure *600 AD - 900 AD*



Details

Los Angeles County Museum of Art



Los Angeles County Museum of Art



Sculpture



Mexico

Share

Compare

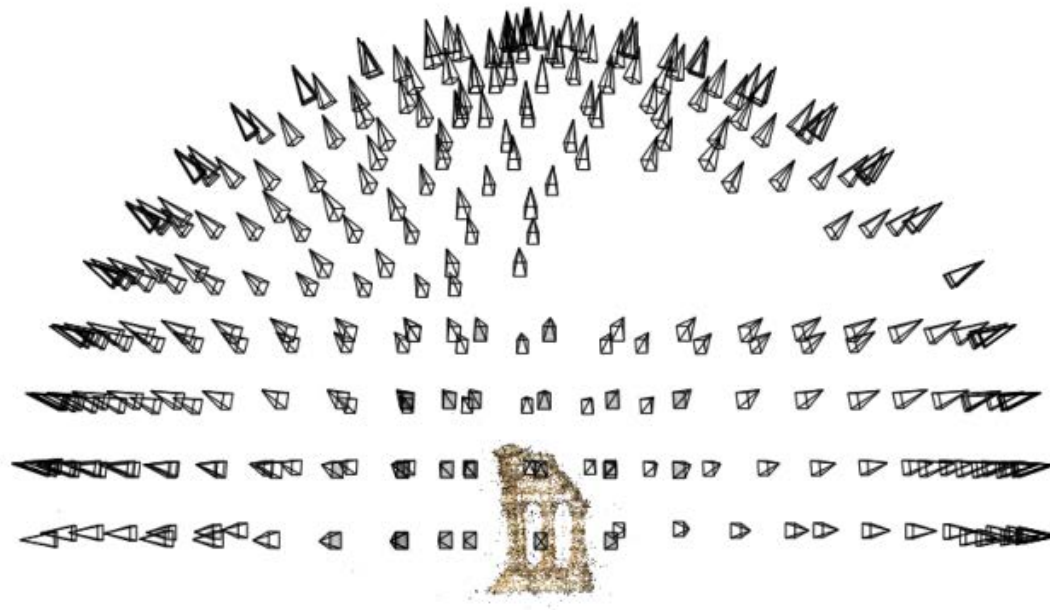
Saved

Discover

Google

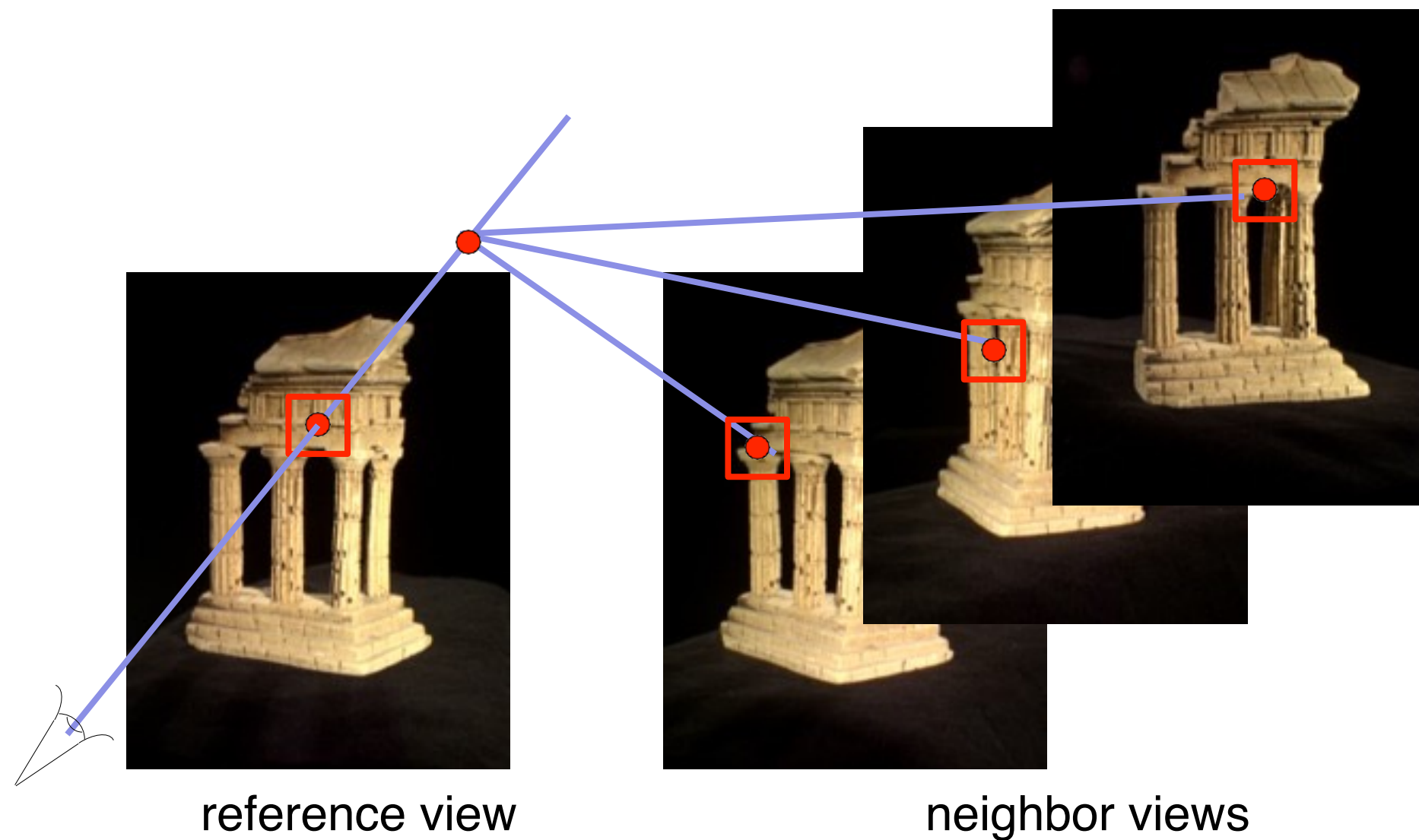
More than two cameras: Multiview Stereo

Multi-view Stereo

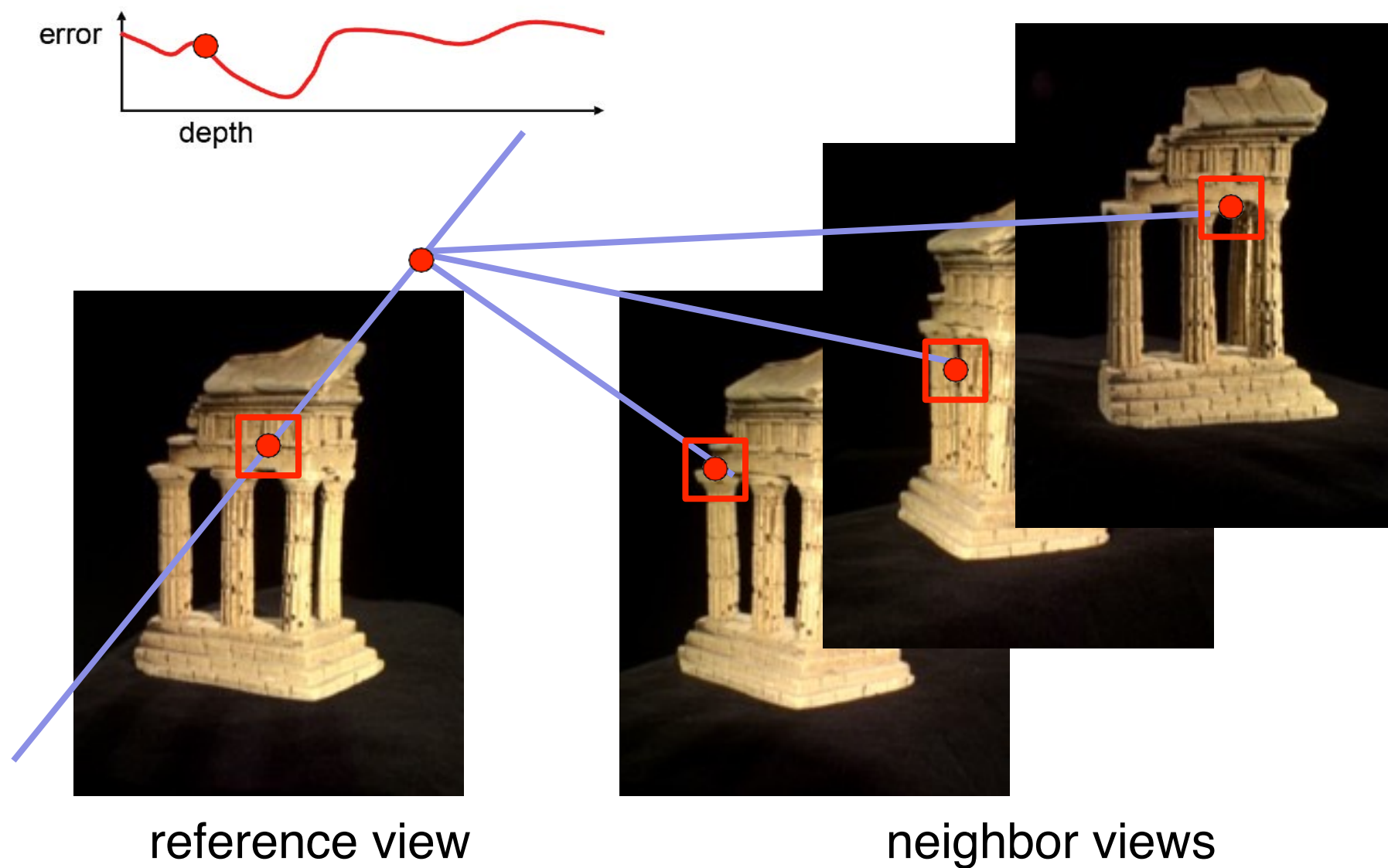


Given several images of the same object or scene, compute a representation of its 3D shape

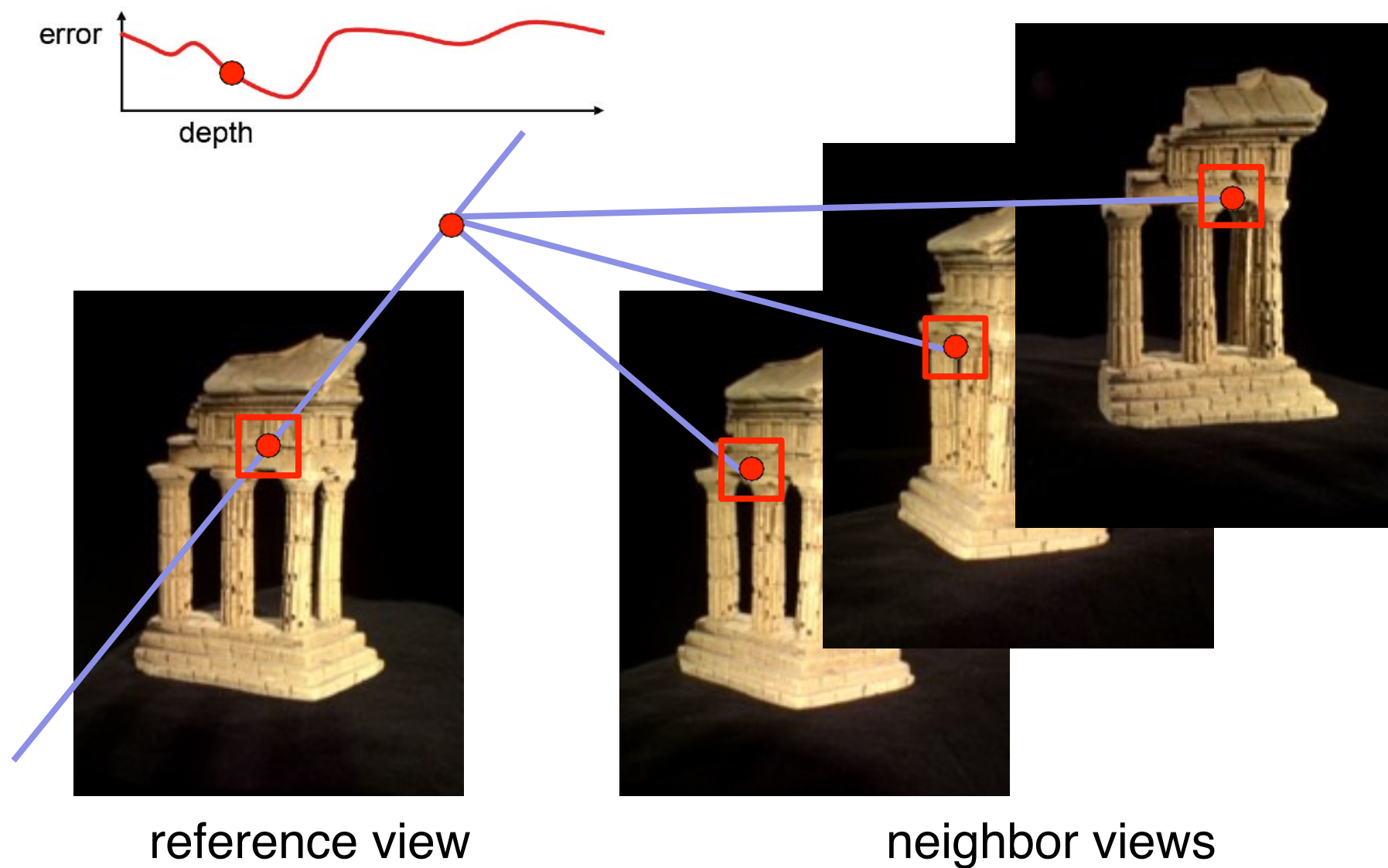
Multi-view stereo: Basic idea



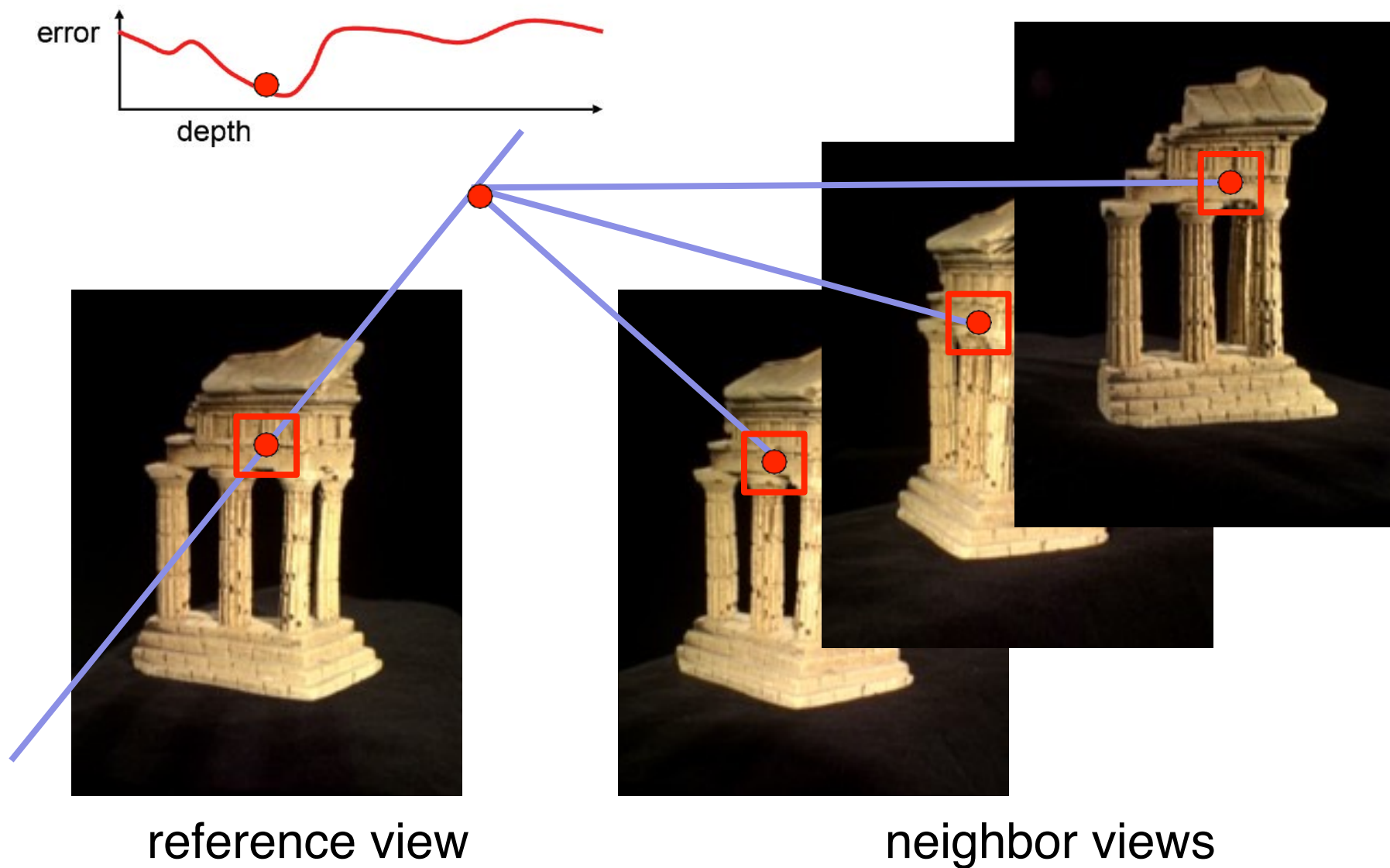
Multi-view stereo: Basic idea



Multi-view stereo: Basic idea



Multi-view stereo: Basic idea



In this manner, solve for a depth map over the whole reference view

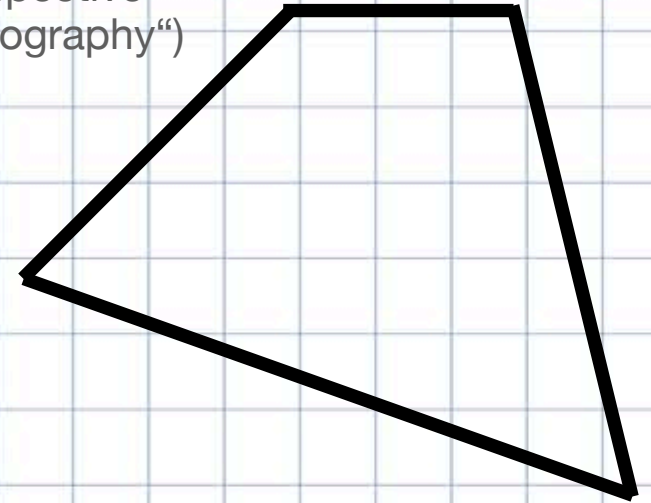
Multi-view stereo: advantages

- Can match windows using more than 1 neighbor, giving a **stronger match signal**
- If you have lots of potential neighbors, can **choose the best subset** of neighbors to match per reference image
- Can reconstruct a depth map for each reference frame, and the merge into a **complete 3D model**

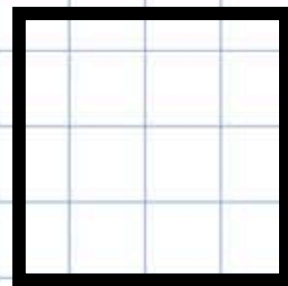
But this is similar to
block matching - not
ideal as we know.

The perspective transformation describes the observation of a plane from different views

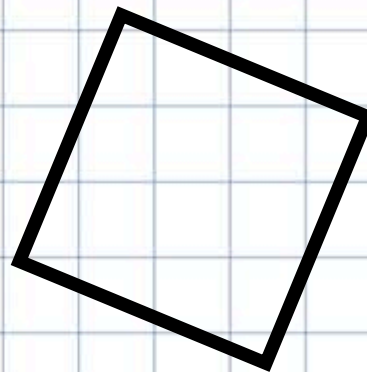
Perspective
(„Homography“)



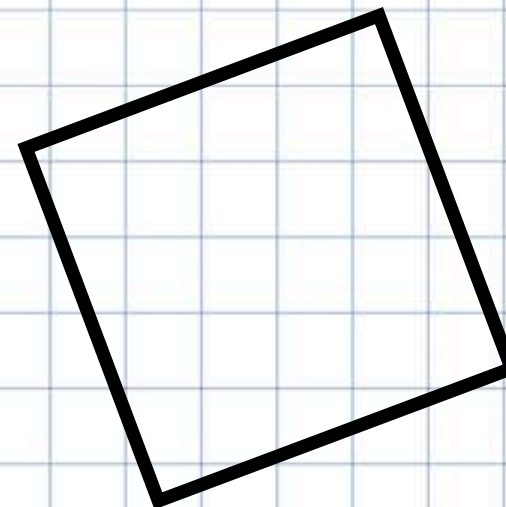
Translation



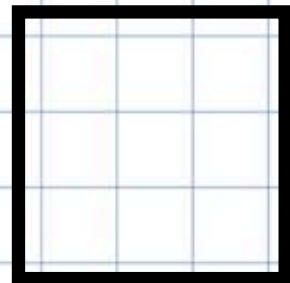
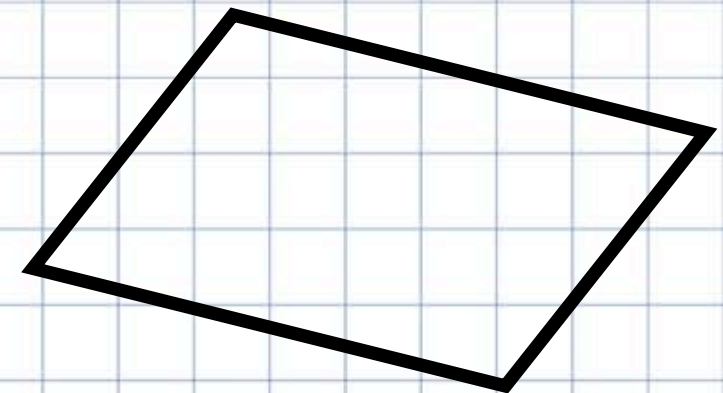
Rigid



Similarity



Affine



Perspective transformation

Using the full 3×3 matrix, we obtain the perspective transformation

$$\tilde{\mathbf{x}}' = \tilde{\mathbf{H}} \cdot \tilde{\mathbf{x}} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \tilde{\mathbf{x}}$$

Corresponding points in two images follow the same transformation, if their 3D projections sit on the same 3D plane!

$\tilde{\mathbf{H}}$ only preserves straight lines.

It requires homogeneous coordinates.

The Euclidean result requires normalization, thus

$$x' = \frac{h_{11}x + h_{12}y + h_{13}w}{h_{31}x + h_{32}y + h_{33}w} \quad \text{and} \quad y' = \frac{h_{21}x + h_{22}y + h_{23}w}{h_{31}x + h_{32}y + h_{33}w}$$

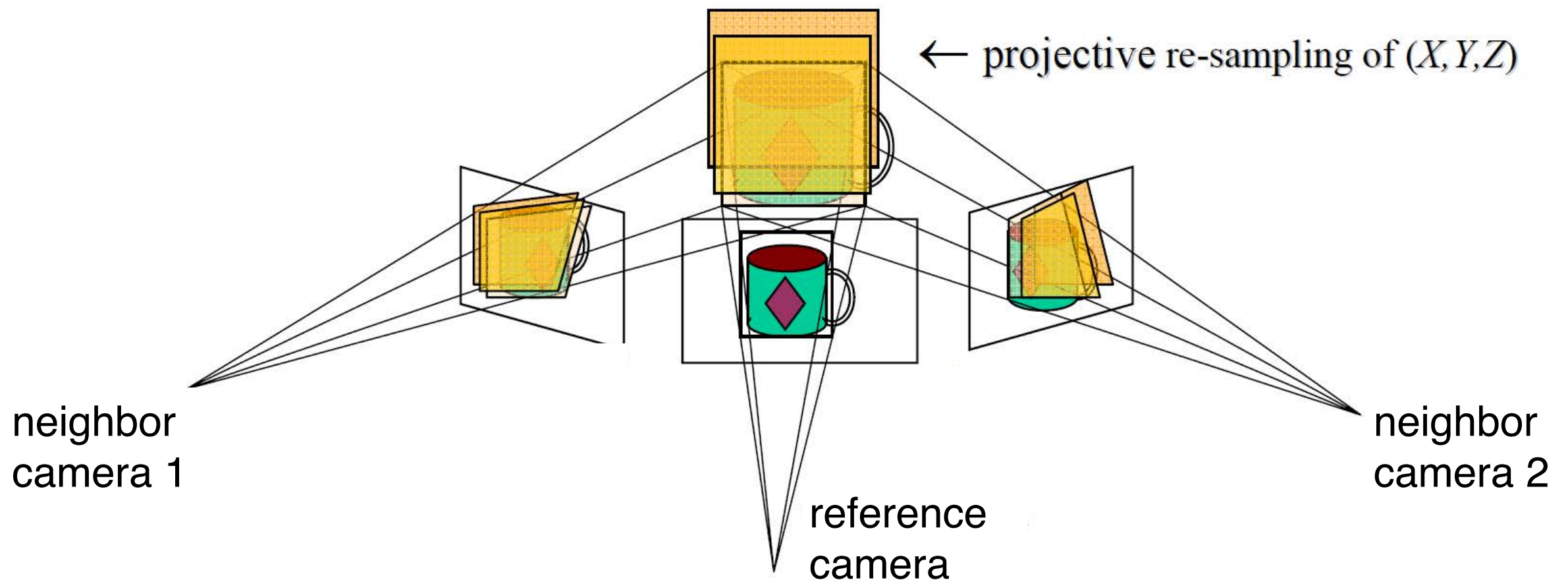
Real-time Plane-sweeping Stereo with Multiple Sweeping Directions

D. Gallup¹, J.-M. Frahm¹, P. Mordohai¹, Q. Yang², and M. Pollefeys¹

¹ Department of Computer Science	² Center for Visualization and Virtual Environments
University of North Carolina	University of Kentucky
Chapel Hill, USA	Lexington, USA

Plane-Sweep Stereo

- Sweep family of planes parallel to the reference camera image plane
- Reproject neighbors onto each plane (via homography) and compare reprojections



Plane-Sweep Stereo



Left neighbor



Reference image



Right neighbor



Left neighbor projected into reference image



Average images on each plane



Right neighbor projected into reference image

Plane-Sweep Stereo



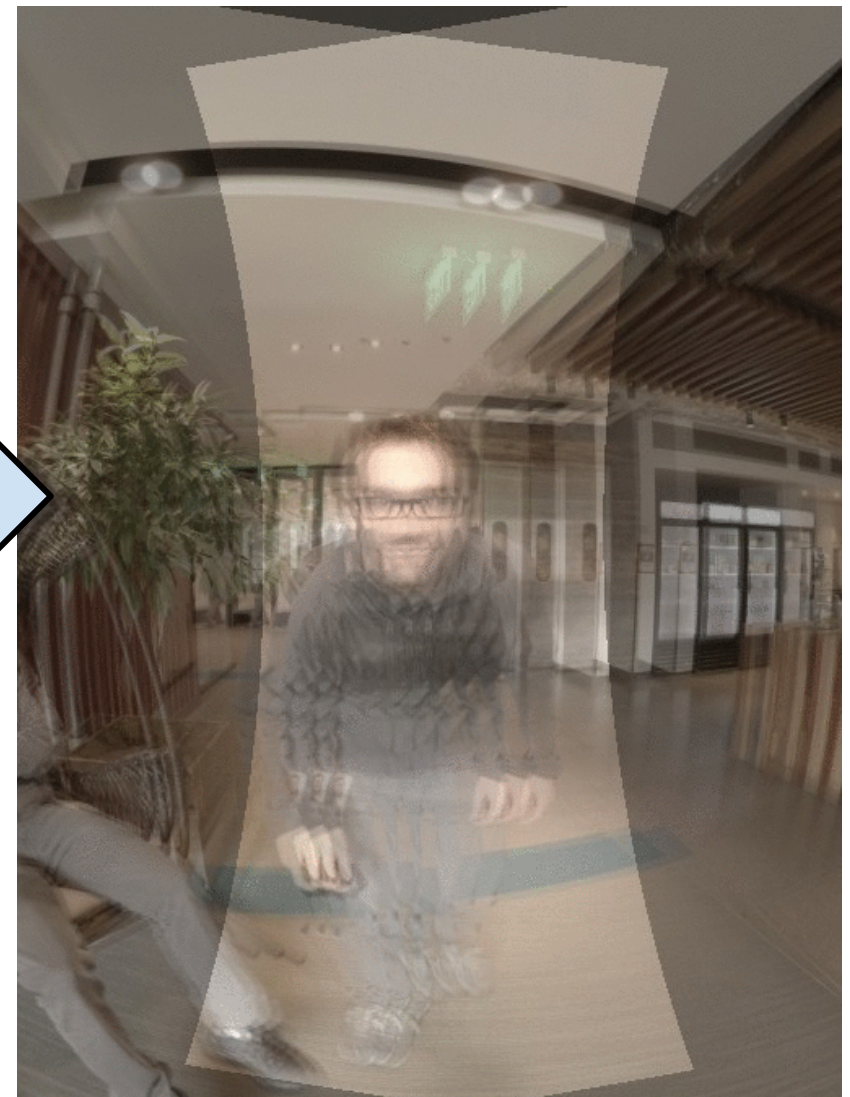
Left neighbor



Reference image



Right neighbor



Planar image reprojections swept over depth
(averaged)

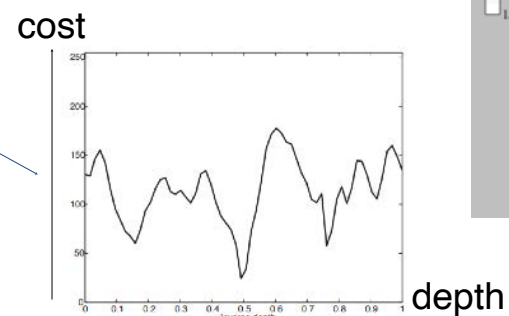
For each pixel, we can compute a cost for each depth of the plane based on the consistency of the observed image patches (similar to template matching in two-view stereo).

From costs to depth maps

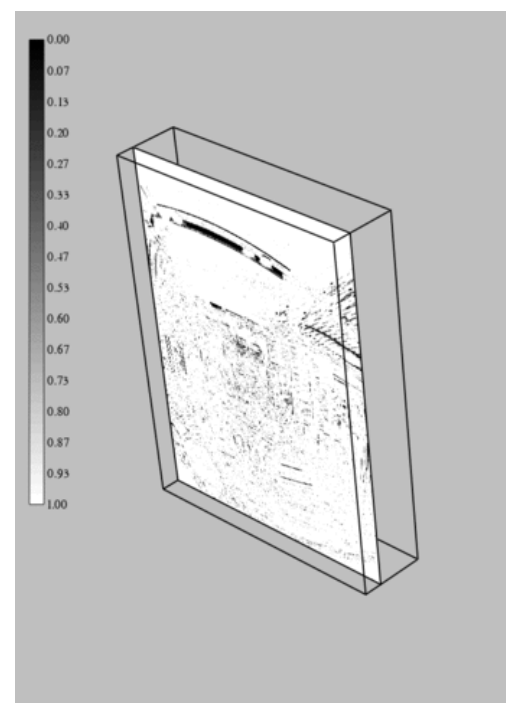


Reference image

Plane sweep



Single pixel's
cost profile



Full cost volume

Depth map
solver

(Belief
propagation,
graph cuts,
etc.)



Depth map

MVSNet (Yao 2018)

